

# Dataset Analysis for Decision-Making

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sicor\_municipal\_co... .csv  
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



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To analyze the dataset and address any specific analytical question, I first need to examine the structure and contents of the `sicor_municipal_contracts.csv` file. Below is a step-by-step breakdown of how I would approach this task:

## Step 1: Load and Inspect the Dataset

I will load the dataset using Python's `pandas` library and inspect its structure to understand its key characteristics.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
```

This code provides:

- Dataset Overview** : Information about column names, data types, and non-null counts.
- Sample Data** : The first few rows of the dataset to understand its structure and content.

## Step 2: Identify Key Characteristics

From the dataset inspection, I will identify:

- The number of rows and columns.
- Key variables (columns) and their potential significance.
- Missing or null values.
- Data types (e.g., numerical, categorical, datetime).
- Any patterns or relationships that could be relevant for analysis.

## Step 3: Address the Analytical Question

Once you provide a specific analytical question, I will use the dataset to answer it. For example:

- If the question involves summarizing contract values by municipality, I will group the data accordingly.
- If the question involves identifying trends over time, I will analyze temporal data (if present).

## Example Analytical Question and Solution

Question:

"What is the total value of contracts awarded in each municipality?"



#### Solution:

Assuming the dataset contains columns like `Municipality` and `Contract_Value`, here's how I would calculate the total contract value per municipality:

```
python
1 # Group by 'Municipality' and sum the 'Contract_Value'
2 contract_totals = data.groupby('Municipality')['Contract_Value'].sum().reset_index()
3
4 # Rename columns for clarity
5 contract_totals.columns = ['Municipality', 'Total_Contract_Value']
6
7 # Display the results
8 print("\nTotal Contract Value by Municipality:")
9 print(contract_totals)
```

This code:

1. Groups the data by the `Municipality` column.
2. Sums the `Contract_Value` for each municipality.
3. Displays the results in a clear format.

#### Deliverables

1. **Clear Answer** : A concise response to the analytical question based on the processed data.
2. **Python Code** : The exact code used to extract, process, and analyze the data.

#### Next Steps

Please provide the specific analytical question you'd like me to address using the dataset. Once I have the question, I can deliver the precise analysis and code.

How elastic is the total contract value in response to changes in the investment area?



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To address the question of how elastic the total contract value is in response to changes in the investment area, we need to calculate and analyze the relationship between these two variables. Elasticity measures the percentage change in one variable (e.g., total contract value) in response to a percentage change in another variable (e.g., investment area).

#### Steps to Analyze Elasticity

##### 1. Understand the Variables :

- Identify the columns corresponding to `Total_Contract_Value` and `Investment_Area`.
- Ensure both variables are numerical and appropriately formatted.

##### 2. Calculate Elasticity :

- Use the formula for elasticity:

$$\text{Elasticity} = \frac{\% \Delta \text{Total Contract Value}}{\% \Delta \text{Investment Area}}$$

- This can be approximated by calculating the ratio of the relative changes in the two variables.

##### 3. Aggregate Data :

- Group the data by `Investment_Area` and calculate the total contract value for each area.
- Compute percentage changes between consecutive levels of the investment area.

##### 4. Interpret Results :

- Determine whether the relationship is elastic ( $>1$ ), inelastic ( $<1$ ), or unitary elastic ( $=1$ ).

#### Python Code Implementation

Below is the Python code to compute and analyze the elasticity:

```
python
1 import pandas as pd
2
3 # Load the dataset
```

```

4 file_path = 'sior_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Check if required columns exist
8 required_columns = {'Investment_Area', 'Contract_Value'}
9 if not required_columns.issubset(data.columns):
10     raise ValueError(f"The dataset must contain the following columns: {required_columns}")
11
12 # Aggregate total contract value by Investment_Area
13 aggregated_data = data.groupby('Investment_Area')['Contract_Value'].sum().reset_index()
14 aggregated_data.columns = ['Investment_Area', 'Total_Contract_Value']
15
16 # Sort by Investment_Area to ensure proper ordering
17 aggregated_data = aggregated_data.sort_values(by='Investment_Area').reset_index(drop=True)
18
19 # Calculate percentage changes
20 aggregated_data['Pct_Change_Investment_Area'] = aggregated_data['Investment_Area'].pct_change()
21 aggregated_data['Pct_Change_Total_Contract_Value'] = aggregated_data['Total_Contract_Value'].pct_change()
22
23 # Calculate elasticity
24 aggregated_data['Elasticity'] = (
25     aggregated_data['Pct_Change_Total_Contract_Value'] / aggregated_data['Pct_Change_Investment_Area']
26 )
27
28 # Drop rows with NaN values (caused by pct_change for the first row)
29 aggregated_data = aggregated_data.dropna()
30
31 # Display results
32 print("\nElasticity Analysis:")
33 print(aggregated_data[['Investment_Area', 'Total_Contract_Value', 'Elasticity']])

```

## Explanation of the Code

### 1. Data Aggregation :

- The dataset is grouped by `Investment_Area`, and the total contract value is summed for each area.

### 2. Percentage Changes :

- The `pct_change()` function computes the percentage change between consecutive rows for both `Investment_Area` and `Total_Contract_Value`.

### 3. Elasticity Calculation :

- Elasticity is computed as the ratio of the percentage change in `Total_Contract_Value` to the percentage change in `Investment_Area`.

### 4. Result Interpretation :

- The output includes the investment area, total contract value, and calculated elasticity for each level of the investment area.

## Example Output

The output will look something like this:

| INVESTMENT_AREA | TOTAL_CONTRACT_VALUE | ELASTICITY |
|-----------------|----------------------|------------|
| 100             | 500000               | 1.2        |
| 200             | 700000               | 0.8        |
| 300             | 900000               | 1.5        |

- An elasticity greater than 1 indicates that the total contract value is highly responsive to changes in the investment area.
- An elasticity less than 1 suggests that the total contract value is less responsive to changes in the investment area.

## Key Insights

- If the elasticity values are consistently above 1, it indicates that the total contract value is elastic with respect to the investment area.
- If the elasticity values are below 1, the relationship is inelastic.
- Fluctuations in elasticity values suggest varying responsiveness at different levels of the investment area.

## Next Steps



If you provide additional context about the dataset (e.g., column names, sample data), I can refine the analysis further.  
Alternatively, if there are specific aspects of elasticity you'd like to explore (e.g., visualization, subgroup analysis), let me know!

